

X1

9/20/82
~\$35

Preliminary

To use the VIO-X in the video mode the board must be set up properly. Please refer to page 5 of your user manual. Switch #1 (the underline enable) must be closed or disabled.

Given that the system is properly set up, you are now ready to test the video board. The test program below was written in basic, so you will need a system disk with basic on it.

- 1) Boot the system.
- 2) Load and run test programs #1, #2, and #3.
- 3) Now read the directions.

A series of characters should have been displayed across the screen in all three tests. Since most people (including myself) hate to read directions we gave you something to do. Now for an over simplified explanation.

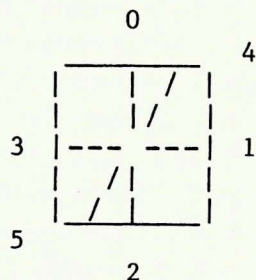
The graphics characters are generated from a 2 by 3 matrix. Each cell (refer to figure 1) of the matrix is called a pixel. The numbering of the pixels are set by the writer of the firmware. With the VIO-X, the lower right pixel is 0 and the lower left is 3.

Figure 1

5	2
4	1
3	0

Lines are generated according to figure 2. For examples of the above and all special characters refer to the appendix.

Figure 2



If you are programming in assembly and want to convert the matrix to binary or hex the following applies. The graphics character takes up one byte. Bit 7 and 6 are reserved for the mode (more about that later). The rest of the bits relate to the pixels directly. If you want to turn on pixel 2 then bit 2 should be "1". See figure 3.

figure 3

7	6	5	4	3	2	1	0	
0	0	1	1	0	0			= 0C Hex
0	1	0	1	1	0			= 16 Hex
1	0	0	1	0	1			= 25 Hex
1	1	1	1	1	1			= 3F Hex

(The first two bits are set according to the mode and will change these values. Refer to your manual.)

Access to the graphics set is divided into three areas we'll call modes. They are:

- 1) Underline
- 2) Underline inverse
- 3) Transparent

The first and second modes are accessed by generating an escape sequence and then the mode code. To exit the video mode you need to generate the escape sequence and the exit code. The test programs demonstrate this.

The escape sequences are as follows:

- 1) Esc Z (= Underline
- 2) Esc Z ' = Underline inverse
- 3) Esc Z 0 = Exit graphics

The transparent mode is entered by:

- 1) Esc U = Transparent mode
- 2) Esc Esc = Exit Transparent mode.

If you read the manual you will see that the transparent mode echoes every character that is entered. When you exit this mode the VIO-X will display two unwanted characters. To get rid of them back space twice and print two blank spaces. This is demonstrated in test program 3. Later revisions of firmware will change this procedure.

Test 1

```

10 PRINT CHR$(26)          'CLEAR SCREEN
15 'TURN ON VIDEO (ESC Z), SET MODE (UNDERLINE)
20 PRINT CHR$(27) + CHR$(90) + CHR$(40)
30 FOR X = 160 TO 255
40   PRINT CHR$(69);      'PRINT DIVIDER BETWEEN CHARACTERS
50   PRINT CHR$(X);       'PRINT VIDEO CHARACTER
60 NEXT X
70 PRINT CHR$(27) + CHR$(90) + CHR$(48) 'EXIT VIDEO MODE
80 PRINT
100 INPUT "AGAIN";A$
110 IF A$ <> "N" THEN 10
120 END

```

Test 2

```

10 PRINT CHR$(26)          'CLEAR SCREEN
20 'TURN ON VIDEO (ESC Z), SET MODE (UNDERLINE INVERSE)
30 PRINT CHR$(27) + CHR$(90) + CHR$(44)
40 FOR X = 160 TO 255
50   PRINT CHR$(63);      'PRINT BLANK AS SEPERATOR
60   PRINT CHR$(X);       'PRINT PRINT VIDEO CHARACTER
70 NEXT X
80 PRINT CHR$(27) + CHR$(90) + CHR$(48) 'EXIT VIDEO MODE
90 PRINT
100 INPUT "AGAIN";A$
110 IF A$ <> "N" THEN 10
120 END

```

Test 3

```

10 PRINT CHR$(26)          'CLEAR SCREEN
20 PRINT CHR$(27) + CHR$(85) 'TURN ON TRANSPARENT MODE
30 FOR X = 128 TO 255
40   PRINT X; CHR$(X);     'PRINT NUMBER, THEN CHARACTER
50 NEXT X
60 PRINT CHR$(27) + CHR$(27); 'CLEAR TRANSPARENT MODE
70 'DOUBLE BACKSPACE THEN PRINT TWO BLANKS TO CLEAR ESC MARKS
80 PRINT CHR$(8) + CHR$(8) + CHR$(32) + CHR$(32)
90 INPUT "AGAIN";A$
100 IF A$ <> "N" THEN 10
110 END

```

Samples of Alternate Programs

Sample 1

```
10 PRINT CHR$(26)
20 FOR X = 128 TO 255
30     PRINT CHR$(27) + CHR$(85) + CHR$(X);
40     PRINT CHR$(27) + CHR$(27);
45     PRINT CHR$(8) + CHR$(8) + CHR$(32) + CHR$(32),
50 NEXT X
60 STOP
200 PRINT CHR$(26)
```

Sample 2

```
10 PRINT CHR$(27) + CHR$(42)
20 INPUT "NUMBER";Z
30 PRINT CHR$(27) + CHR$(61) + CHR$(40) + CHR$(40);
40 PRINT CHR$(27) + CHR$(85); CHR$(Z);
50 PRINT CHR$(27) + CHR$(27)
60 PRINT CHR$(27) + CHR$(61) + CHR$(40) + CHR$(41) + CHR$(32) + CHR$(32)
70 GOTO 20
```

Sample 3

```
10 PRINT CHR$(26)
20 FOR X = 160 TO 255
30     PRINT CHR$(27) + CHR$(90) + CHR$(40) + CHR$(X);
40     PRINT CHR$(27) + CHR$(90) + CHR$(48),
50 NEXT X
60 STOP
```


CHR\$(172)

```

. . . . .
.* * * * *
.* * * * *
.* * * * *
.* * * * *
.
.* * * *
.* * * *
.* * * *
.* * * *
. . . . .

```

CHR\$(173)

```

. . . . .
.* * * * *
.* * * * *
.* * * * *
.* * * * *
.
.* * * * *
.* * * * *
.* * * * *
.* * * * *
. . . . .

```

CHR\$(174)

```

. . . . .
.* * * * *
.* * * * *
.* * * * *
.* * * * *
.      * * * *
.      * * * *
.* * * * *
.* * * *
.* * * *
.* * * *
. . . . .

```

CHR\$(175)

```

. . . . .
.* * * * *
.* * * * *
.* * * * *
.* * * * *
.      * * * *
.      * * * *
.* * * * *
.* * * * *
.* * * * *
.* * * * *
. . . . .

```

CHR\$(176)

```

. . . . .
.* * * *
.* * * *
.* * * *
.* * * *
.* * * *
.* * * *
.* * * *
.
.
.
.
. . . . .

```

CHR\$(177)

```

. . . . .
.* * * *
.* * * *
.* * * *
.* * * *
.* * * *
.* * * *
.* * * * *
.      * * * *
.      * * * *
.      * * * *
. . . . .

```

CHR\$(178)

```

. . . . .
.* * * *
.* * * *
.* * * *
.* * * * *
.* * * * *
.* * * * *
.* * * * *
.
.
.
.
. . . . .

```

CHR\$(179)

```

. . . . .
.* * * *
.* * * *
.* * * *
.* * * * *
.* * * * *
.* * * * *
.* * * * *
.      * * * *
.      * * * *
.      * * * *
. . . . .

```

CHR\$(180)

```

. . . . .
.* * * * *
.* * * * *
.* * * * *
.* * * * *
.* * * *
.* * * *
.* * * *
.
.
.
.
. . . . .

```

CHR\$(181)

```

. . . . .
.* * * * *
.* * * * *
.* * * * *
.* * * * *
.* * * *
.* * * *
.* * * * *
.      * * * *
.      * * * *
.      * * * *
. . . . .

```

CHR\$(182)

```

. . . . .
.* * * * *
.* * * * *
.* * * * *
.* * * * *
.* * * * *
.* * * * *
.* * * * *
.
.
.
.
. . . . .

```

CHR\$(183)

```

. . . . .
.* * * * *
.* * * * *
.* * * * *
.* * * * *
.* * * * *
.* * * * *
.* * * * *
.      * * * *
.      * * * *
.      * * * *
. . . . .

```

A 10x10 grid of dots. In the center, there is a 4x4 square of asterisks. The asterisks are located at the intersections of the 4th, 5th, 6th, and 7th rows and the 4th, 5th, 6th, and 7th columns (assuming 1-indexing from the top-left).

A 10x10 grid of dots. In the top-left corner, there is a 4x4 square of asterisks. The asterisks are located at rows 1-4 and columns 1-4. All other positions in the grid are occupied by dots.

A 10x10 grid of dots. In the center, there is a 4x4 subgrid of stars. The stars are located at the intersections of the 4th, 5th, 6th, and 7th rows and the 4th, 5th, 6th, and 7th columns (assuming 1-indexing from the top-left).

A 10x10 grid of dots. In the center, there is a 6x6 square of asterisks. The asterisks are located at the intersections of the 3rd to 8th rows and the 3rd to 8th columns (using 1-based indexing from the top-left). The dots form the outer border and fill the remaining space within the 10x10 grid.

A 10x10 grid of asterisks. In the top-left corner, a 3x3 square of dots is overlaid, covering the first three rows and the first three columns. The dots are located at positions (row, column) where both row and column are 1, 2, or 3. All other positions in the 10x10 grid contain an asterisk.

```

      .   .   .   .   .   .   .
      *   *   *   *   *
      *   *   *   *   *
      *   *   *   *   *
      *   *   *   *   *
      *   *   *   *   *
      .   .   .   .   .   .   .

```

A 10x10 grid of dots. The dots are arranged in 10 rows and 10 columns. In the center of the grid, there are four asterisks (*) arranged in a 2x2 square pattern. Specifically, the asterisks are located at the intersections of the 5th and 6th rows with the 5th and 6th columns (assuming the top-left dot is at row 1, column 1).

A 10x10 grid of dots. The dots are arranged in a 10x10 square. The number '10' is formed by asterisks. The '1' is a vertical column of 5 asterisks in the 4th column from the left. The '0' is a horizontal row of 4 asterisks in the 5th row from the top, starting from the 4th column and extending to the 7th column.

The diagram consists of a large rectangle formed by small dots. The dots are arranged in a grid-like fashion along the perimeter. Inside this rectangular boundary, there are seven larger asterisk symbols (*) scattered at various positions.

A 10x10 grid of dots. A path of asterisks starts at the center (row 5, column 5) and moves towards the top-right corner. The path consists of the following coordinates (row, column): (5, 5), (5, 6), (5, 7), (5, 8), (5, 9), (6, 9), (7, 9), (8, 9), (9, 9), (9, 10).

A 10x10 grid of dots. A vertical line of 10 asterisks is located in the 5th column from the left. A horizontal line of 10 asterisks is located in the 5th row from the top. The asterisks intersect at the dot in the 5th row and 5th column.

A 10x10 grid of dots. Asterisks are placed at the following coordinates (row, column) starting from (0,0) at the top-left:

- Row 1: (1,4), (1,5), (1,6), (1,9)
- Row 2: (2,4), (2,5), (2,6), (2,9)
- Row 3: (3,4), (3,5), (3,6), (3,9)
- Row 4: (4,4), (4,5), (4,6), (4,9)
- Row 5: (5,4), (5,5), (5,6), (5,9)
- Row 6: (6,4), (6,5), (6,6), (6,9)
- Row 7: (7,1), (7,2), (7,3), (7,4), (7,5), (7,6), (7,9)
- Row 8: (8,1), (8,2), (8,3), (8,4), (8,5), (8,6), (8,9)
- Row 9: (9,1), (9,2), (9,3), (9,4), (9,5), (9,6), (9,9)
- Row 10: (10,1), (10,2), (10,3), (10,4), (10,5), (10,6), (10,9)

A 10x10 grid of dots. Asterisks are placed at the following coordinates (row, column) starting from (0,0) at the top-left:

- Row 1: (1,4), (1,5), (1,6), (1,9)
- Row 2: (2,4), (2,5), (2,6), (2,9)
- Row 3: (3,4), (3,5), (3,6), (3,9)
- Row 4: (4,4), (4,5), (4,6), (4,9)
- Row 5: (5,4), (5,5), (5,6), (5,9)
- Row 6: (6,1), (6,2), (6,3), (6,4), (6,5), (6,6), (6,7), (6,8)
- Row 7: (7,4)
- Row 8: (8,4)
- Row 9: (9,1), (9,2), (9,3), (9,4), (9,5), (9,6), (9,7), (9,8)

CHR\$(220)

```

. . . . .
.           *
.         *
.       *
.     *
.   *
. * * * *
.   *
.   *
.   *
.   *
. . . . .

```

CHR\$(221)

```

. . . . .
.       *       *
.     * *
.   * *
. * *
. * *
. * *
. * *
. * *
. *
. . . . .

```

CHR\$(222)

```

. . . . .
.           *
.         *
.       *
.     *
.   *
. * * * *
.   *
.   *
.   *
.   *
. . . . .

```

CHR\$(223)

```

. . . . .
.       *       *
.     * *
.   * *
. * *
. * *
. * *
. * *
. * *
. *
. . . . .

```

CHR\$(224)

```

. . . . .
.
.
.
.
.
.   *
. *
. *
. *
. *
. . . . .

```

CHR\$(225)

```

. . . . .
.       *
.       *
.       *
.       *
.       *
.   *
. *
. *
. *
. *
. . . . .

```

CHR\$(226)

```

. . . . .
.
.
.
.
.
. * * * *
. *
. *
. *
. *
. . . . .

```

CHR\$(227)

```

. . . . .
.       *
.       *
.       *
.       *
.       *
.   * * * *
. *
. *
. *
. *
. . . . .

```

CHR\$(228)

```

. . . . .
.
.
.
.
.
.   *
. * *
. *
. *
. *
. *
. . . . .

```

CHR\$(229)

```

. . . . .
.       *
.       *
.       *
.       *
.       *
.   *
. *
. *
. *
. *
. *
. . . . .

```

CHR\$(230)

```

. . . . .
.
.
.
.
.
. * * * *
. *
. *
. *
. *
. *
. . . . .

```

CHR\$(231)

```

. . . . .
.       *
.       *
.       *
.       *
.       *
.   * * * *
. *
. *
. *
. *
. *
. . . . .

```

CHR\$(208)

CHR\$ 209)

CHR\$ 210)

CHR\$ 211)

CHR\$(212)

CHR\$(213)

CHR\$(214)

CHR\$(215)

CHR\$(216)

CHR\$(217)

CHR\$(218)

CHR\$(219)

CHR\$(232)

```

. . . . .
.
.
.
.
. * * * *
.   *
.   *
.   *
. *
. . . . .

```

CHR\$(233)

```

. . . . .
.       *
.       *
.       *
.       *
.       *
. * * * *
.   *
.   *
.   *
. *
. . . . .

```

CHR\$(234)

```

. . . . .
.
.
.
.
. * * * * *
.   *
.   *
.   *
. *
. . . . .

```

CHR\$ 235)

```

. . . . .
.       *
.       *
.       *
.       *
.       *
. * * * * *
.   *
.   *
.   *
. *
. . . . .

```

CHR\$(236)

```

. . . . .
.
.
.
.
. * * * *
.   * *
.   * *
.   * *
. *
. . . . .

```

CHR\$(237)

```

. . . . .
.       *
.       *
.       *
.       *
.       *
. * * * *
.   * *
.   * *
.   * *
. *
. . . . .

```

CHR\$(238)

```

. . . . .
.
.
.
.
. * * * * *
.   * *
.   * *
.   * *
. *
. . . . .

```

CHR\$(239)

```

. . . . .
.       *
.       *
.       *
.       *
.       *
. * * * * *
.   * *
.   * *
.   * *
. *
. . . . .

```

CHR\$(240)

```

. . . . .
.       *
.       *
.       *
.       *
.       *
. * * * *
.   *
.   *
.   *
. *
. . . . .

```

CHR\$(241)

```

. . . . .
.       *
.       *
.       *
.       *
.       *
. * * * *
.   *
.   *
.   *
. *
. . . . .

```

CHR\$(242)

```

. . . . .
.       *
.       *
.       *
.       *
.       *
. * * * *
.   *
.   *
.   *
. *
. . . . .

```

CHR\$(243)

```

. . . . .
.       *
.       *
.       *
.       *
.       *
. * * * *
.   *
.   *
.   *
. *
. . . . .

```


CHR\$(172)

```

. . . . .
.   * * * *.
.   * * * *.
.   * * * *.
.   * * * *.
.
.
.
. * * * *
. * * * *
. * * * *
. * * * *
. . . . .

```

CHR\$(173)

```

. . . . .
.   * * * *.
.   * * * *.
.   * * * *.
.   * * * *.
.
.
.
. * * * * * * *
. * * * * * * *
. * * * * * * *
. * * * * * * *
. . . . .

```

CHR\$(174)

```

. . . . .
.   * * * *.
.   * * * *.
.   * * * *.
.   * * * *.
.   * * * *.
.   * * * *.
. * * * * * * *
. * * * *
. * * * *
. * * * *
. . . . .

```

CHR\$(175)

```

. . . . .
.   * * * *.
.   * * * *.
.   * * * *.
.   * * * *.
.   * * * *.
.   * * * *.
. * * * * * * *
. * * * * * * *
. * * * * * * *
. * * * * * * *
. . . . .

```

CHR\$(176)

```

. . . . .
.
.
.
. * * * *
. * * * *
. * * * *
. * * * *
.
.
.
. . . . .

```

CHR\$(177)

```

. . . . .
.
.
.
. * * * *
. * * * *
. * * * *
. * * * * * * *
.
.   * * * *.
.   * * * *.
.   * * * *.
. . . . .

```

CHR\$(178)

```

. . . . .
.
.
.
. * * * * * * *
. * * * * * * *
. * * * * * * *
. * * * * * * *
.
.
.
. . . . .

```

CHR\$(179)

```

. . . . .
.
.
.
. * * * * * * *
. * * * * * * *
. * * * * * * *
. * * * * * * *
.   * * * *.
.   * * * *.
.   * * * *.
. . . . .

```

CHR\$(180)

```

. . . . .
.   * * * *.
.   * * * *.
.   * * * *.
. * * * * * * *
. * * * *
. * * * *
. * * * *
.
.
.
. . . . .

```

CHR\$(181)

```

. . . . .
.   * * * *.
.   * * * *.
.   * * * *.
. * * * * * * *
. * * * *
. * * * *
. * * * * * * *
.
.   * * * *.
.   * * * *.
.   * * * *.
. . . . .

```

CHR\$(182)

```

. . . . .
.   * * * *.
.   * * * *.
.   * * * *.
. * * * * * * *
. * * * * * * *
. * * * * * * *
. * * * * * * *
.
.
.
. . . . .

```

CHR\$(183)

```

. . . . .
.   * * * *.
.   * * * *.
.   * * * *.
. * * * * * * *
. * * * * * * *
. * * * * * * *
. * * * * * * *
.   * * * *.
.   * * * *.
.   * * * *.
. . . . .

```


A 10x10 grid of dots. In the center, there is a 4x4 subgrid of asterisks. The asterisks are located at rows 3-6 and columns 3-6 (assuming 0-indexing from the top-left dot).

A 10x10 grid of dots. The top-left 4x4 square of the grid contains asterisks (*), while the remaining positions are empty dots.

A 10x10 grid of dots. The dots are arranged in a pattern that resembles a stylized 'C' or a bracket on the left side. The first column has 10 dots. The second column has 9 dots, missing the top one. The third column has 8 dots, missing the top two. The fourth column has 7 dots, missing the top three. The fifth column has 6 dots, missing the top four. The sixth column has 5 dots, missing the top five. The seventh column has 4 dots, missing the top six. The eighth column has 3 dots, missing the top seven. The ninth column has 2 dots, missing the top eight. The tenth column has 1 dot, missing the top nine.

A 10x10 grid of dots. The central 6x6 area is filled with asterisks, forming a solid square. The outer border of the grid consists of dots, with the four corners also being dots.

A 10x10 grid of dots. In the center, there is a 6x6 subgrid of stars. The stars are located at the intersections of the 4th through 9th rows and the 3rd through 8th columns (using 1-based indexing from the top-left).

A 10x10 grid of dots. The top-left 3x3 square of dots is missing, leaving a 3x3 square of empty space. All other positions in the 10x10 grid contain a dot.

A 10x10 grid of dots. The dots are arranged in a regular pattern. Some dots are replaced by asterisks (*). The asterisks are located at the following positions (row, column) starting from the top-left corner:

- (1, 2), (1, 3), (1, 4), (1, 7), (1, 8)
- (2, 1), (2, 2), (2, 3), (2, 4), (2, 7), (2, 9)
- (3, 1), (3, 2), (3, 3), (3, 4), (3, 7), (3, 9), (3, 10)
- (4, 1), (4, 2), (4, 3), (4, 7), (4, 9), (4, 10)
- (5, 1), (5, 2), (5, 3), (5, 7), (5, 8), (5, 9), (5, 10)
- (6, 1), (6, 2), (6, 3), (6, 7), (6, 8), (6, 9), (6, 10)
- (7, 1), (7, 2), (7, 3), (7, 4), (7, 7), (7, 8), (7, 9), (7, 10)
- (8, 1), (8, 2), (8, 3), (8, 4), (8, 7), (8, 8), (8, 9), (8, 10)
- (9, 1), (9, 2), (9, 3), (9, 4), (9, 7), (9, 8), (9, 9), (9, 10)
- (10, 1), (10, 2), (10, 3), (10, 4), (10, 7), (10, 8), (10, 9), (10, 10)

A 15x15 grid of dots. The dots are arranged in a square pattern. Some dots are replaced by asterisks (*). The asterisks form a pattern that is mostly solid but has several missing dots, notably in the top right and middle right areas.

A 15x15 grid of dots. The dots are arranged in a pattern that is roughly rectangular, spanning from row 2 to row 14 and column 2 to column 14. The dots are replaced by asterisks in a specific pattern. The asterisks are located at the following (row, column) positions: (2,2), (2,3), (2,4), (2,5), (2,6), (2,14), (3,2), (3,3), (3,4), (3,5), (3,6), (3,14), (4,2), (4,3), (4,4), (4,5), (4,6), (4,14), (5,2), (5,3), (5,4), (5,5), (5,6), (5,14), (6,2), (6,3), (6,4), (6,5), (6,6), (6,14), (7,2), (7,3), (7,4), (7,5), (7,6), (7,14), (8,2), (8,3), (8,4), (8,5), (8,6), (8,14), (9,2), (9,3), (9,4), (9,5), (9,6), (9,14), (10,2), (10,3), (10,4), (10,5), (10,6), (10,14), (11,2), (11,3), (11,4), (11,5), (11,6), (11,14), (12,2), (12,3), (12,4), (12,5), (12,6), (12,14), (13,2), (13,3), (13,4), (13,5), (13,6), (13,14), (14,2), (14,3), (14,4), (14,5), (14,6), (14,14).

```

. . . . .
.*.*.*. . *.*.
.*.*.*. *.*.
.*.*.*. *.*.
.*.*.*.  *.*.
.*.*.*.  *.*.
.  *.*.*.
.*.*.*.*.*.*.
.*.*.*.*.*.*.
.*.*.*.*.*.*.
.*.*.*.*.*.*.
. . . . .

```

A 10x10 grid of dots. The dots are arranged in a pattern that resembles a stylized letter 'A' or a similar shape. The pattern is composed of several connected blocks of dots. The top row has 10 dots. The second row has 9 dots, with the last dot missing. The third row has 8 dots, with the last two dots missing. The fourth row has 7 dots, with the last three dots missing. The fifth row has 6 dots, with the last four dots missing. The sixth row has 5 dots, with the last five dots missing. The seventh row has 4 dots, with the last six dots missing. The eighth row has 3 dots, with the last seven dots missing. The ninth row has 2 dots, with the last eight dots missing. The tenth row has 1 dot, with the last nine dots missing.

CHR\$(232)

```

. . . . .
.*.*.*.*.*.
.*.*.*.*.*.
.*.*.*.*.*.
.*.*.*.*.*.
.*.*.*.*.*.
.   .   .   .
.*.*   .   .
.*   .   .   .
.*   .   .   .
.   .   .   .
. . . . .

```

CHR\$(233)

```

. . . . .
.*.*.*   .
.*.*.*   .
.*.*.*   .
.*.*.*   .
.*.*.*   .
.   .   .   .
.*.*   .   .
.*   .   .   .
.*   .   .   .
.   .   .   .
. . . . .

```

CHR\$(234)

```

. . . . .
.*.*.*.*.*.
.*.*.*.*.*.
.*.*.*.*.*.
.*.*.*.*.*.
.*.*.*.*.*.
.   .   .   .
.*.*   .   .
.*   .   .   .
.*   .   .   .
.   .   .   .
. . . . .

```

CHR\$(235)

```

. . . . .
.*.*.*   .
.*.*.*   .
.*.*.*   .
.*.*.*   .
.*.*.*   .
.   .   .   .
.*.*   .   .
.*   .   .   .
.*   .   .   .
.   .   .   .
. . . . .

```

CHR\$(236)

```

. . . . .
.*.*.*.*.*.
.*.*.*.*.*.
.*.*.*.*.*.
.*.*.*.*.*.
.*.*.*.*.*.
.   .   .   .
.*.*   .   .
.*   .   .   .
.*   .   .   .
.   .   .   .
. . . . .

```

CHR\$(237)

```

. . . . .
.*.*.*   .
.*.*.*   .
.*.*.*   .
.*.*.*   .
.*.*.*   .
.   .   .   .
.*.*   .   .
.*   .   .   .
.*   .   .   .
.   .   .   .
. . . . .

```

CHR\$(238)

```

. . . . .
.*.*.*.*.*.
.*.*.*.*.*.
.*.*.*.*.*.
.*.*.*.*.*.
.*.*.*.*.*.
.   .   .   .
.*.*   .   .
.*   .   .   .
.*   .   .   .
.   .   .   .
. . . . .

```

CHR\$(239)

```

. . . . .
.*.*.*   .
.*.*.*   .
.*.*.*   .
.*.*.*   .
.*.*.*   .
.   .   .   .
.*.*   .   .
.*   .   .   .
.*   .   .   .
.   .   .   .
. . . . .

```

CHR\$(240)

```

. . . . .
.*.*.*.*.*.
.*.*.*.*.*.
.*.*.*.*.*.
.*.*.*.*.*.
.*.*.*.*.*.
.*.*.*.*.*.
.*.*.*.*.*.
.*.*.*.*.*.
.*.*.*.*.*.
.   .   .   .
. . . . .

```

CHR\$(241)

```

. . . . .
.*.*.*   .
.*.*.*   .
.*.*.*   .
.*.*.*   .
.*.*.*   .
.*.*.*   .
.*.*.*   .
.*.*.*   .
.*.*.*   .
.   .   .   .
. . . . .

```

CHR\$(242)

```

. . . . .
.*.*.*.*.*.
.*.*.*.*.*.
.*.*.*.*.*.
.*.*.*.*.*.
.*.*.*.*.*.
.*.*.*.*.*.
.*.*.*.*.*.
.*.*.*.*.*.
.*.*.*.*.*.
.   .   .   .
. . . . .

```

CHR\$(243)

```

. . . . .
.*.*.*   .
.*.*.*   .
.*.*.*   .
.*.*.*   .
.*.*.*   .
.*.*.*   .
.*.*.*   .
.*.*.*   .
.*.*.*   .
.   .   .   .
. . . . .

```

CHR \$(248)

CHR\$ (249)

CHR\$ (250)

CHR\$ (251)

CHR\$(252)

CHR\$ (253)

CHR\$ (254)

[illegible]

CHR\$(255)

1

CHR\$ (131)

CHR\$ (135)

CHR\$ (139)

A 10x10 grid of dots. In the center, there is a 5x5 sub-grid of asterisks. The asterisks are located at the intersections of the 4th, 5th, 6th, 7th, and 8th rows and the 4th, 5th, 6th, 7th, and 8th columns (assuming 1-indexing from the top-left).

CHR\$(140)

```

. . . . .
.
.
.
.
. * * * * *
. * * * * *
.
.
.
.
. . . . .

```

CHR\$(141)

```

. . . . .
. * * *
. * * *
. * * *
. * * *
. * * *
. * * *
. * * *
. * * *
. * * *
. . . . .

```

CHR\$(142)

```

. . . . .
. * * *
. * * *
. * * *
. * * *
. * * *
. * * *
. * * *
.
.
.
.
. . . . .

```

CHR\$(143)

```

. . . . .
. * * *
. * * *
. * * *
. * * *
. * * *
. * * *
. * * *
.
.
.
.
. . . . .

```

CHR\$(144)

```

. . . . .
.
.
.
.
.
.
.
.
. * * * * *
. * * * * *
. . . . .

```

CHR\$(145)

```

. . . . .
.
.
.
.
.
.
. * * * * *
. * * * * *
. * * * * *
. . . . .

```

CHR\$(146)

```

. . . . .
.
.
.
.
. * * * * *
. * * * * *
. * * * * *
. * * * * *
. * * * * *
. * * * * *
. * * * * *
. . . . .

```

CHR\$(147)

```

. . . . .
.
.
.
. * * * * *
. * * * * *
. * * * * *
. * * * * *
. * * * * *
. * * * * *
. * * * * *
. . . . .

```

CHR\$(148)

```

. . . . .
. * * * * *
. * * * * *
.
.
.
.
.
.
.
.
. . . . .

```

CHR\$(149)

```

. . . . .
. * * * * *
. * * * * *
. * * * * *
. * * * * *
.
.
.
.
.
.
. . . . .

```

CHR\$(150)

```

. . . . .
. * * * * *
. * * * * *
. * * * * *
. * * * * *
. * * * * *
. * * * * *
.
.
.
.
. . . . .

```

CHR\$(151)

```

. . . . .
. * * * * *
. * * * * *
. * * * * *
. * * * * *
. * * * * *
. * * * * *
. * * * * *
.
.
.
.
. . . . .

```


CHR\$(152)

```

. . . . .
.*
.*
.*
.*
.*
.*
.*
.*
.*
.*
. . . . .

```

CHR\$(153)

```

. . . . .
.* * *
.* * *
.* * *
.* * *
.* * *
.* * *
.* * *
.* * *
.* * *
. . . . .

```

CHR\$(154)

```

. . . . .
.* * * * *
.* * * * *
.* * * * *
.* * * * *
.* * * * *
.* * * * *
.* * * * *
.* * * * *
.* * * * *
. . . . .

```

CHR\$(155)

```

. . . . .
.*
.*
.*
.*
.*
.*
.*
.*
.*
.*
. . . . .

```

CHR\$(156)

```

. . . . .
.* * *.
.* * *.
.* * *.
.* * *.
.* * *.
.* * *.
.* * *.
.* * *.
.* * *.
. . . . .

```

CHR\$(157)

```

. . . . .
.* * * * *.
.* * * * *.
.* * * * *.
.* * * * *.
.* * * * *.
.* * * * *.
.* * * * *.
.* * * * *.
.* * * * *.
. . . . .

```

CHR\$(158)

```

. . . . .
.* *
.*
.*
.*
.*
.*
.*
.*
.*
. . . . .

```

CHR\$(159)

```

. . . . .
.* *
.*
.*
.*
.*
.*
.*
.*
.*
. . . . .

```

CHR\$(160)

```

. . . . .
.
.
.
.
.
.
.
.
.
.
. . . . .

```

CHR\$(161)

```

. . . . .
.
.*
.*
.*
.*
.
. . . . .

```

CHR\$(162)

```

. . . . .
.
.*
.*
.*
.*
. . . . .

```

CHR\$(163)

```

. . . . .
.*
.*
.*
.*
.*
.*
.*
.*
. . . . .

```


CHR\$(188)

```

. . . . .
.      *
.    *
.  *
. *
. *
.  *
.    *
.      *
. . . . .

```

CHR\$(189)

```

. . . . .
.      *
.      *
.      *
.      *
.      *
.      *
.      *
.      *
. . . . .

```

CHR\$(190)

```

. . . . .
.      *
.    *
.  *
. *
. *
.  *
.    *
.      *
. . . . .

```

CHR\$(191)

```

. . . . .
.      *
.    *
.  *
. *
. *
.  *
.    *
.      *
. . . . .

```

CHR\$(192)

```

. . . . .
.      *
.    *
.  *
. *
. *
.  *
.    *
.      *
. . . . .

```

CHR\$(193)

```

. . . . .
.      *
.    *
.  *
. *
. *
.  *
.    *
.      *
. . . . .

```

CHR\$(194)

```

. . . . .
.      *
.    *
.  *
. *
. *
.  *
.    *
.      *
. . . . .

```

CHR\$(195)

```

. . . . .
.      *
.    *
.  *
. *
. *
.  *
.    *
.      *
. . . . .

```

CHR\$(196)

```

. . . . .
.      *
.    *
.  *
. *
. *
.  *
.    *
.      *
. . . . .

```

CHR\$(197)

```

. . . . .
.      *
.    *
.  *
. *
. *
.  *
.    *
.      *
. . . . .

```

CHR\$(198)

```

. . . . .
.      *
.    *
.  *
. *
. *
.  *
.    *
.      *
. . . . .

```

CHR\$(199)

```

. . . . .
.      *
.    *
.  *
. *
. *
.  *
.    *
.      *
. . . . .

```

A 10x10 grid of dots. In the center, there is a 3x3 square of asterisks. The asterisks are located at the following coordinates (row, column) starting from (0,0) at the top-left: (3,2), (3,3), (3,4), (3,5), (3,6), (4,2), (4,3), (4,4), (4,5), (4,6), (5,2), (5,3), (5,4), (5,5), (5,6), (6,2), (6,3), (6,4), (6,5), (6,6).

[illegible]

A 10x10 grid of dots. Asterisks are placed at the following coordinates (row, column) starting from (0,0) at the top-left:

Row	Col 1	Col 2	Col 3	Col 4	Col 5	Col 6	Col 7	Col 8	Col 9
0									
1									
2		*						*	
3		*						*	
4		*	*					*	
5		*		*				*	
6		*			*			*	
7		*				*		*	
8		*						*	
9		*						*	

A 10x10 grid of dots. Inside this grid, there is a 6x6 square of asterisks. The asterisks are located at positions (row, column) where both row and column are between 2 and 7 (inclusive), forming a solid 6x6 block.

A 10x10 grid of dots. In the center, there is an 8x8 square of asterisks. The asterisks are arranged in a grid that is 2 units from the top and bottom edges and 2 units from the left and right edges of the dot grid.

A 10x10 grid of dots. Asterisks are placed at the following coordinates (row, column) starting from (0,0) at the top-left:

- Row 1: (1,2), (1,3), (1,4)
- Row 2: (2,1), (2,2), (2,9), (2,10)
- Row 3: (3,1), (3,2), (3,9), (3,10)
- Row 4: (4,1), (4,2), (4,5), (4,9), (4,10)
- Row 5: (5,1), (5,2), (5,7), (5,9), (5,10)
- Row 6: (6,3), (6,4), (6,9)

[illegible]

A 10x10 grid of dots. Asterisks are placed at the following coordinates (row, column) starting from (0,0) at the top-left:

- Row 1: (1,1), (1,4), (1,5), (1,6), (1,9)
- Row 2: (2,1), (2,2), (2,3), (2,4), (2,5), (2,6), (2,7), (2,8), (2,9)
- Row 3: (3,1), (3,2), (3,3), (3,4), (3,5), (3,6), (3,7), (3,8), (3,9)
- Row 4: (4,1), (4,2), (4,3), (4,4), (4,5), (4,6), (4,7), (4,8), (4,9)
- Row 5: (5,1), (5,2), (5,3), (5,4), (5,5), (5,6), (5,7), (5,8), (5,9)
- Row 6: (6,1), (6,2), (6,3), (6,4), (6,5), (6,6), (6,7), (6,8), (6,9)
- Row 7: (7,1), (7,2), (7,3), (7,4), (7,5), (7,6), (7,7), (7,8), (7,9)
- Row 8: (8,1), (8,2), (8,3), (8,4), (8,5), (8,6), (8,7), (8,8), (8,9)
- Row 9: (9,1), (9,2), (9,3), (9,4), (9,5), (9,6), (9,7), (9,8), (9,9)

A 10x10 grid of dots. In the center, there is a 3x3 subgrid of asterisks. The asterisks are located at the intersections of the 4th, 5th, and 6th rows and the 4th, 5th, and 6th columns (assuming the top-left dot is at row 1, column 1).

A 10x10 grid of dots. In the center, there is a 3x3 subgrid of asterisks. The asterisks are located at the following coordinates (row, column) starting from (0,0) at the top-left:

Row	Col 1	Col 2	Col 3	Col 4	Col 5	Col 6
3		*	*	*		
4	*	*	*	*	*	*
5	*	*	*	*		

A 10x10 grid of dots. The dots are arranged in 10 rows and 10 columns. The following positions (row, column) contain an asterisk (*): (3, 4), (3, 5), (3, 8), (4, 2), (4, 3), (4, 7), (4, 8), (5, 2), (5, 3), (5, 7), (5, 8), (6, 2), (6, 3), (6, 7), (6, 8), (7, 4), (7, 5), (8, 4), (8, 5), (8, 6), (8, 7), (8, 8), (8, 9), (9, 2), (9, 3), (9, 4), (9, 5), (9, 6), (9, 7), (9, 8), (9, 9).

A 10x10 grid of dots. In the center column, there are five asterisks (*) arranged vertically, centered between the horizontal midline and the top and bottom edges.

A 10x10 grid of dots. The number '10' is formed by asterisks. The '1' is a vertical column of 6 asterisks in the 7th column from the left, starting from the 4th row from the top. The '0' is formed by asterisks at the 3rd, 4th, 5th, 6th, 8th, and 9th rows from the top, in the 3rd and 9th columns from the left.

CHR\$(236)

```

. . . . .
.
. *
. *
. *
. *
. *
. *
. *
.
.
.
. . . . .

```

CHR\$(237)

```

. . . . .
.
.
.
. * *
. * * *
. * * *
. * * *
. * *
.
.
.
. . . . .

```

CHR\$(238)

```

. . . . .
.
.
.
. * * *
. * * *
. * * *
. * * *
. * *
.
.
.
. . . . .

```

CHR\$(239)

```

. . . . .
.
.
.
. * * *
. * * *
. * * *
. * * *
. * * *
.
.
.
. . . . .

```

CHR\$(240)

```

. . . . .
.
.
.
. * * *
. * * *
. * * *
. * * *
. * * *
. *
.
.
. . . . .

```

CHR\$(241)

```

. . . . .
.
.
.
. * * *
. * * *
. * * *
. * * *
. * * *
. *
.
.
. . . . .

```

CHR\$(242)

```

. . . . .
.
.
.
. * * *
. * * *
. * * *
. * * *
. *
.
.
. . . . .

```

CHR\$(243)

```

. . . . .
.
.
.
. * * * *
. * * * *
. * * * *
. * * * *
. * * * *
.
.
.
. . . . .

```

CHR\$(244)

```

. . . . .
.
.
.
. *
. *
. * * *
. *
. *
. *
. *
.
.
.
. . . . .

```

CHR\$(245)

```

. . . . .
.
.
.
. *
. *
. *
. *
. *
. *
. *
.
.
.
. . . . .

```

CHR\$(246)

```

. . . . .
.
.
.
. *
. *
. *
. *
. *
. *
. *
.
.
.
. . . . .

```

CHR\$(247)

```

. . . . .
.
.
.
. *
. *
. *
. *
. *
. *
. *
.
.
.
. . . . .

```


CHR\$(248)

```

. . . . .
.
.
.
. *      *
.  *    *
.   *  *
.  *    *
. *      *
.
.
.
.

```

CHR\$(249)

```

. . . . .
.
.
.
. *      *
.  *    *
.   *  *
.  *    *
. *      *
.   * *
. * * *
.

```

CHR\$(250)

```

. . . . .
.
.
.
. * * * *
.   *
.     *
. * * * *
.
.
.
.

```

CHR\$(251)

```

. . . . .
.
.
.
. * *
.   *
.   *
. * *
.   *
.   *
. * *
.

```

CHR\$(252)

```

. . . . .
.
. *
. *
. *
. *
. *
. *
. *
. *
. *
.

```

CHR\$(253)

```

. . . . .
.
.
. * *
.   *
.   *
. * *
.   *
. *
.

```

CHR\$(254)

```

. . . . .
.
.
.
. *
. *
. *
.

```

CHR\$(255)

```

. . . . .
. * * * *
. * * * *
. * * * *
. * * * *
. * * * *
. * * * *
. * * * *
. * * * *
.

```